

Differentiable Flowsheets for Rare Earth Separations

Gradient-Based Process Design and Optimization

John Kitchen

Carnegie Mellon University

The Challenge: Rare Earth Separation

- U.S. depends on imports for **critical REE** (Nd, Dy, Pr) used in magnets, electronics, defense
- REE are **chemically near-identical** — separation requires many stages of solvent extraction
- Traditional simulators (Aspen, HYSYS) lack efficient optimization for these flowsheets
- Process design today relies on **trial-and-error** experimentation and heuristics

Separating 10+ similar elements across 20+ stages with recycles is a combinatorial nightmare for conventional tools.

Our Approach: Differentiable Flowsheets

- Built on **JAX** — every calculation is automatically differentiable
- Gradients flow through the **entire flowsheet**: units, recycles, economics
- Gradient-based optimization: **orders of magnitude faster** than derivative-free methods
- Key insight: treat the whole process as a **differentiable program**

Lab Data → **Calibrated Model** → **Optimized Design** → **Economics**

All connected by automatic differentiation

What "Differentiable" Means in Practice

Traditional Approach

- Perturb one variable, re-simulate
- **N+1 simulations** for N parameters
- Finite-difference approximations
- Slow, inaccurate for large systems

Our Approach

- **Exact gradients** of any output w.r.t. any input — in one pass
- Sensitivity, uncertainty, and optimization simultaneously
- Scales to hundreds of parameters

Example: $\frac{\partial(\text{Nd recovery})}{\partial(\text{pH})} = 1.43$ — raising pH by 0.1 increases Nd recovery by 14.3%

REE Separation Capabilities

Elements & Extractants

- **10 REE elements:** La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Y
- **4 extractant systems:** D2EHPA, PC88A, Cyanex272, TBP
- pH-dependent equilibrium with temperature correction

Unit Operations

- Multi-stage **Extractor** (Kremser equation)
- **Mixer-Settler** with efficiency
- **Scrubber** for impurity rejection
- **Stripper** for product recovery
- **3 Precipitators** (oxalate, carbonate, hydroxide)
- **CeO₂ oxidizer** for selective Ce removal

Pre-Built Flowsheet Templates

Template	Configuration	Application
ExtractStripCircuit	2-section	Basic separations
ExtractScrubStripCircuit	3-section	High-purity products
SplitShellCascade	Multi-product	Multiple pure REE streams
FullSeparationTrain	Complete	Ore-to-product processing
GroupSeparator	Group split	Light / Middle / Heavy REE

All templates are **differentiable end-to-end** — optimize any parameter with gradients.

Application: NdFeB Magnet Production

Feed & Design

- Feed: La 34%, **Nd 45%**, Pr 11%, Dy 2%
- Extractant: **PC88A** selected automatically (SF 4.03 vs D2EHPA 2.66)
- pH optimized via gradient descent → **pH 5.0**

Circuit

- 8 extraction + 6 scrubbing + 4 stripping stages

Results

- **100% Nd recovery**
- **83% product purity**
- 82% Pr recovery

Sensitivity (automatic)

- $\partial(\text{recovery})/\partial(\text{pH})$ — continuous
- Effect of adding stages quantified

Integrated Economics — All Differentiable

500 t/year Plant

- **CAPEX**: \$3.6M
- **OPEX**: \$10.3M/year
- **Revenue**: \$48.1M/year
- **ROI**: 774%
- **Payback**: < 1 year

What Differentiable Economics Enables

- Optimize directly for **profit** or **minimum selling price**
- Gradients of cost w.r.t. every design variable
- Trade-off: purity vs. recovery vs. cost — quantified exactly
- Risk-aware design under price uncertainty

Parameter Estimation from Lab Data

- Fit pH-dependent distribution coefficients to **experimental concentration data**
- 7 pH values × 3 replicates → **9 parameters** fitted (a, b, c per element)
- All parameters recovered within **95% confidence intervals**
- **$R^2 > 0.999$** , RMSE < 1 mg/L

*Closes the loop: **lab data** → **calibrated model** → **optimized design***

No manual parameter tuning required.

Uncertainty Quantification

Four Built-In Methods

- **Linear propagation** — fast, first-order
- **Monte Carlo** — nonparametric sampling
- **Sobol indices** — global sensitivity
- **Covariance propagation** — full Jacobian

Impact

- Propagate experimental uncertainty through flowsheet to **economic outputs**
- Identify which parameters matter most
- **Risk-aware optimization**: maximize expected profit while controlling variance

What Makes This Novel

- **First differentiable REE separation simulator**
- Exact gradients through multi-stage counter-current extraction with recycles
- **Unified framework**: experiment → model → optimize → economics
- Three domain plugins: REE extraction, bio manufacturing, carbon capture
- Open, extensible Python framework — not a black-box commercial tool

Traditional simulators **simulate**. Our framework **optimizes**.

Current Status & Next Steps

Demonstrated

- 10 REE elements, 4 extractants
- 5 flowsheet templates
- NdFeB production optimization
- Parameter estimation with UQ
- Full technoeconomic analysis

Next Steps

- Pilot plant validation
- Additional extractant systems
- Multi-objective optimization (purity vs. cost vs. recovery)
- Lab-to-plant design acceleration

Vision: *A computational tool that takes lab extraction data and delivers an optimized, costed plant design — automatically.*

Thank You

Questions?

John Kitchin — Carnegie Mellon University

`diffflow` — Differentiable Flowsheet Framework